

The geometric visualization of a vector such as $\begin{bmatrix} 3 \\ -1 \end{bmatrix}$ is often aided by including an arrow (directed line segment) from the origin $(0,0)$ to the point $(3,-1)$, as in Figure 2.

The following rule can be verified by analytic geometry.

Parallelogram Rule for Addition

If u and v in $\mathbb{R}^2$ are represented as points in the plane, then $u+v$ corresponds to the fourth vertex of the parallelogram whose other vertices are u , 0 and v . See Figure 3.

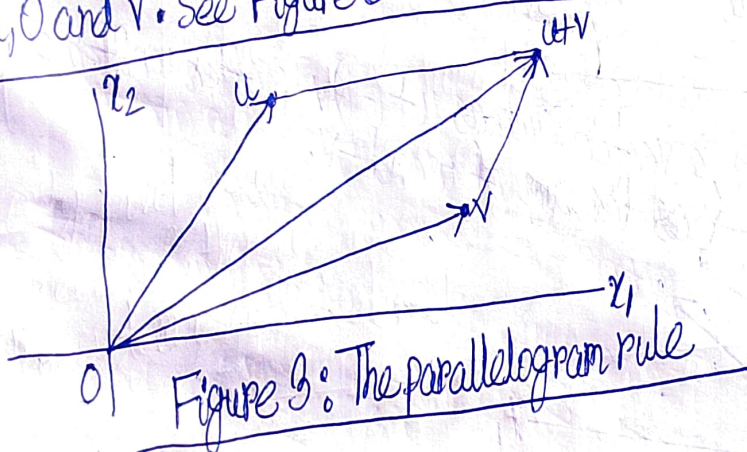


Figure 3: The parallelogram rule

Vectors in $\mathbb{R}^3$

Vectors in $\mathbb{R}^3$ are 3×1 column matrices with three entries. They are represented geometrically by points in a three-dimensional coordinate space with arrows from the origin, sometimes included for visual clarity.

Vectors in $\mathbb{R}^n$

If n is a positive integer, $\mathbb{R}^n$ (read "r-n") denotes the collection of all lists (or ordered n -tuples) of n real numbers, usually written as $n \times 1$ column matrices, such as

$$u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$$

The vector whose entries are all zero is called the zero vector, and is denoted by 0 .

Equality of vectors in $\mathbb{R}^n$ and the operations of scalar multiplication and vector addition in $\mathbb{R}^n$ are defined entry by entry just as in $\mathbb{R}^2$.

These operations on vectors have the following properties, which can